LESSON PLAN

BASIC RADIO & AVIONICS (25)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction, need for starter, ignition, brief description of a typical engine starting system
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction, need for starter, ignition, brief description of a typical engine starting system. Practical on Engine Starting system.
5. Specific Objectives	25.1.2.1 Trainee will be able to know and understand the constructional details of starters, ignition units and principles of. operation of various engine instruments

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Practical on Engine Starting system; Safety precautions: Ensure engine is cold, verify fuel flow, check battery voltage; Instructor supervision: Monitor trainee actions during start sequence and verify instrument readings.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an engine starter?
(b) Why is an ignition system necessary for engine operation?
(c) List three instruments found on a typical engine starting panel.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to engine starting systems • Need for starter in aircraft engines • Basic principles of ignition systems • Components of a typical starting system • Sequence of engine start procedure	L & Ds	PPT, IB	15 min
(b)	• Constructional details of starter motors • Operation of ignition units • Instrument principles and indications • System interconnections and controls • Safety considerations during start	L & Ds	PPT, PDF	20 min
(c)	• Practical demonstration of engine start • Verification of instrument readings • Troubleshooting common start faults • Review of start sequence steps • Documentation of start parameters	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the role of the starter motor during the initial engine rotation phase.
(b) Describe how the ignition unit ensures reliable combustion during start.

12. Home Assignment and Follow Up	Prepare a brief note on the differences between electric starter motors and cartridge turbo starters for review in LP 05.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Brief description of engine starter motor, electric starter motor, cartridge turbo starter. Types of starting system, auxiliary power unit, air starter and battery trolleys Purpose and principle of operation of high energy ignition unit	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min